

Magic Square

In any magic square, the first number i.e. 1 is stored at position $(n/2, n-1)$. Let this position be (i, j) .

The next number is stored at position $(i-1, j+1)$ where we can consider each row & column as circular array i.e. they wrap around.

2	7	6
9	5	1
4	3	8

$$\text{Sum} = n * (n^2 + 1) / 2 = 30 / 2 = 15$$

Magic Square

2		
		1
	3	

1. position of number 1 = $(3/2, 3-1) = (1, 2)$

2. position of number 2 = $(1-1, 2+1) = (0, 0)$

3. position of number 3 = $(0-1, 0+1) = (3-1, 1) = (2, 1)$

The position of next number is calculated by decrementing row number of the previous number by 1, and incrementing the column number of the previous number by 1.

At any time, if the calculated row position becomes -1, it will wrap around to n-1. Similarly, if the calculated column position becomes n, it will wrap around to 0.

Magic Square

2		6
	5	1
4	3	

4. position of number 4 = $(2-1, 1+1) = (1, 2)$
Since, at this position, 1 is there.
So, apply condition 2.
new position = $(1+1, 2-2) = (2, 0)$

5. position of number 5 = $(2-1, 0+1) = (1, 1)$

6. position of number 6 = $(1-1, 1+1) = (0, 2)$

If the magic square already contains a number at the calculated position, calculated column position will be decremented by 2, and calculated row position will be incremented by 1.

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2	7	6
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7. position of number 7 = $(0-1, 2+1) = (-1, 3)$

This is tricky,

see condition 3 new position = $(0, 3-2) = (0, 1)$

8. position of number 8 = $(0-1, 1+1) = (-1, 2) = (2, 2)$ //wrap around

9. position of number 9 = $(2-1, 2+1) = (1, 3) = (1, 0)$ //wrap around

If the **calculated row position** is **-1** & **calculated column position** is **n**, the new position would be: **$(0, n-2)$** .

Magic Square

Three conditions hold:

1. The position of next number is calculated by decrementing row number of the previous number by 1, and incrementing the column number of the previous number by 1.

At any time, if the calculated row position becomes -1, it will wrap around to $n-1$. Similarly, if the calculated column position becomes n , it will wrap around to 0.

2. If the magic square already contains a number at the calculated position, calculated column position will be decremented by 2, and calculated row position will be incremented by 1.

3. If the calculated row position is -1 & calculated column position is n , the new position would be: $(0, n-2)$.

C Program

```
void generateSquare(int n)
{
    int magicSquare[n][n];
    int i,j,num;
    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
            magicSquare[i][j]=0;
    }
    i = n / 2;
    j = n - 1;
```

```
for (i = 0; i < n; i++)
{
    for (j = 0; j < n; j++)
        printf("%d",magicSquare[i][j]);

    printf("\n");
}
}
```

```
for (num = 1; num <= n * n;)
{
    if (i == -1 && j == n) // 3rd condition
    {
        j = n - 2;
        i = 0;
    }
    else
    {
        if (j == n)
            j = 0;
        if (i < 0)
            i = n - 1;
    }
    if (magicSquare[i][j]) // 2nd condition
    {
        j -= 2;
        i++;
        continue;
    }
    else
        magicSquare[i][j] = num++; // set number
    j++;
    i--; // 1st condition
}
```

```
void main()
{
    int n = 3;
    generateSquare(n);
}
```

Magic Square

9	3	22	16	15
2	21	20	14	8
25	19	13	7	1
18	12	6	5	24
11	10	4	23	17